

# CIRB 2023 Detailed Report - Analysis 1 - Updated

## Analysis 1 = Permit Populations + Geomap

*Any question regarding the notebook, please contact Yung Nguyen or Robert Ford*

*Last Updated: 01/27/2025*

- CIRB data: 2023 Detailed Report
- Started development on 10/01/2024
- Finished extraction on -ongoing-

```
In [1]: #Libraries import
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

import geopandas
from shapely.geometry import Point
import pgeocode
import plotly.graph_objects as go
from geopy.geocoders import Nominatim
from geopy.exc import GeocoderTimedOut
from geopandas import GeoDataFrame
import time
```

```
In [2]: df = pd.read_csv("CIRB2023_detailedreport_total.csv", sep='\t')
```

```
C:\Users\YNguyen\AppData\Local\Temp\ipykernel_25940\704197640.py:1: Dtype
Warning: Columns (23) have mixed types. Specify dtype option on import or
set low_memory=False.
```

```
df = pd.read_csv("CIRB2023_detailedreport_total.csv", sep='\t')
```

## Data Exploration

```
In [3]: df.info()
print(df.columns.tolist())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 374764 entries, 0 to 374763
Data columns (total 28 columns):
#   Column                Non-Null Count  Dtype
---  -
0   COUNTY*               374764 non-null  object
1   CITY*                 374764 non-null  object
2   MONTH*                374764 non-null  object
3   YEAR*                 374764 non-null  float64
4   ISSUE DATE*           374764 non-null  object
5   PERMIT NO              374764 non-null  object
6   SOURCE*                374764 non-null  object
7   CODE*                 374764 non-null  object
8   CLASS*                374764 non-null  object
9   APN                    374749 non-null  object
10  ADDRESS                374764 non-null  object
11  CITY                   374764 non-null  object
12  STATE                  374764 non-null  object
13  ZIP                    374764 non-null  object
14  BUILDER                374752 non-null  object
15  SQ FOOTAGE             374764 non-null  float64
16  UNITS                  374764 non-null  float64
17  VALUATION*             374764 non-null  float64
18  NAME OF OWNER          374763 non-null  object
19  PERMIT COST             374764 non-null  float64
20  NUMBER OF STORIES      374764 non-null  float64
21  CAPACITY               374764 non-null  object
22  DESC                   374689 non-null  object
23  CODE                   374764 non-null  object
24  UCODE_TYPE             374764 non-null  object
25  UCODE_BLG_CAT          374764 non-null  object
26  UCODE_NAME             374764 non-null  object
27  UCODE_DESC             374764 non-null  object
dtypes: float64(6), object(22)
memory usage: 80.1+ MB
['COUNTY*', 'CITY*', 'MONTH*', 'YEAR*', 'ISSUE DATE*', 'PERMIT NO', 'SOURCE*', 'CODE*', 'CLASS*', 'APN', 'ADDRESS', 'CITY', 'STATE', 'ZIP', 'BUILDER', 'SQ FOOTAGE', 'UNITS', 'VALUATION*', 'NAME OF OWNER', 'PERMIT COST', 'NUMBER OF STORIES', 'CAPACITY', 'DESC', 'CODE', 'UCODE_TYPE', 'UCODE_BLG_CAT', 'UCODE_NAME', 'UCODE_DESC']
```

```
In [4]: # General Analysis
len(df)
df['CODE*'].unique()
df.head()
```

Out[4]:

|   | COUNTY* | CITY*                       | MONTH*  | YEAR*  | ISSUE DATE* | PERMIT NO  | SOURCE*   | COD |
|---|---------|-----------------------------|---------|--------|-------------|------------|-----------|-----|
| 0 | BUTTE   | BIGGS                       | JANUARY | 2023.0 | 1/26/2023   | B19-4115   | SECONDARY | U1  |
| 1 | BUTTE   | BUTTE COUNTY UNINCORPORATED | JANUARY | 2023.0 | 1/26/2023   | B19-4115   | SECONDARY | U1  |
| 2 | BUTTE   | CHICO                       | JANUARY | 2023.0 | 1/1/2023    | B22-00934  | PRIMARY   | U1  |
| 3 | BUTTE   | PARADISE                    | MARCH   | 2023.0 | 3/15/2023   | BP22-01671 | PRIMARY   | U1  |
| 4 | BUTTE   | PARADISE                    | APRIL   | 2023.0 | 4/14/2023   | BP23-00012 | SECONDARY | U1  |

5 rows × 28 columns

**Count relevant rows with heat pump in the description**

```
In [77]: heatpump = ['Heat Pump Installation', 'Heat Pump Replacement',
                    'HVAC Heat Pump System', 'Heat Pump Unit', 'Heat Pump Equipmen',
                    'Mini-Split Heat Pump System', 'Ductless Heat Pump',
                    'Central Heat Pump System', 'Packaged Heat Pump Unit', 'Split-Residential Heat Pump',
                    'Domestic Hot Water Heat Pump', 'Heat Hybrid Heat Pump System', 'hpwh', 'vrf',
                    'Variable Refrigerant Flow', 'High Efficiency Heat Pump',
                    'SEER2', 'hp', 'heat pump', 'heatpump', 'electrification', 'hpwh']
heatpump = [x.lower() for x in heatpump]
hp_df = []
for word in heatpump:
    count_hp = df.DESC.str.count(word).sum()
    hp_df.append([word, count_hp])
hp_df = pd.DataFrame(hp_df, columns=['HP_term', 'Frequency'])
```

```
In [78]: hp_df
```

```
Out[78]:
```

|    | HP_term                      | Frequency |
|----|------------------------------|-----------|
| 0  | heat pump installation       | 23.0      |
| 1  | heat pump replacement        | 16.0      |
| 2  | hvac heat pump system        | 0.0       |
| 3  | heat pump unit               | 49.0      |
| 4  | heat pump equipment          | 2.0       |
| 5  | mini-split heat pump system  | 24.0      |
| 6  | ductless heat pump           | 33.0      |
| 7  | central heat pump system     | 5.0       |
| 8  | packaged heat pump unit      | 4.0       |
| 9  | split-system heat pump       | 3.0       |
| 10 | residential heat pump        | 1.0       |
| 11 | domestic hot water heat pump | 0.0       |
| 12 | heat pump water heater       | 30.0      |
| 13 | hybrid heat pump system      | 0.0       |
| 14 | hpwh                         | 0.0       |
| 15 | vrf                          | 0.0       |
| 16 | variable refrigerant flow    | 0.0       |
| 17 | high efficiency heat pump    | 3.0       |
| 18 | seer2                        | 5.0       |
| 19 | hp                           | 137.0     |
| 20 | heat pump                    | 1859.0    |
| 21 | heatpump                     | 31.0      |
| 22 | electrification              | 0.0       |
| 23 | hpwhs                        | 0.0       |

```
In [79]: hp_df.Frequency.sum()
```

```
Out[79]: 2225.0
```

### ***Natural Language Processing: Bag of words***

```
In [58]: from wordcloud import WordCloud, STOPWORDS, ImageColorGenerator
from nltk.tokenize import word_tokenize
import string
from collections import Counter
```

```
In [45]: alldes = df['DESC'].dropna()
alldes = alldes.tolist()
```

```
In [46]: #Tokenize the list
token_alldes = [word_tokenize(i) for i in alldes]
```

```
In [48]: flat_alldes = [item for sublist in token_alldes for item in sublist]
```

```
In [49]: useless_words = list(STOPWORDS) + list(string.punctuation)
```

```
In [51]: clean_flat = [w for w in flat_alldes if not w in useless_words and len(w)
clean_flat = [word for line in clean_flat for word in line.split('.')]
```

```
Out[51]: ['Lane-senior', 'senior']
```

```
In [62]: #Test
sorted(w for w in set(clean_flat) if 'hp' in w)
```

```
Out[62]: ['3hp',
'Enhphase-5P',
'Oshpd',
'Southpoint',
'Southpointe',
'highpile',
'hp',
'hp/ah',
'hpsf']
```

```
In [63]: #Test
sorted(w for w in set(clean_flat) if 'heatpump' in w)
```

```
Out[63]: ['airhandler/heatpump', 'heatpump', 'heatpump/electric', 'heatpumps']
```

```
In [76]: #Test
sorted(w for w in set(clean_flat) if 'electric' in w)
```

```
Out[76]: ['-electrical',
'Gas/electric',
'Gas/electrical',
'HVAC/electrical',
'all-electric',
'cabinets/plumbing/electrical',
'electric',
'electric/plumbing',
'electric/plumbing/mech',
'electrica',
'electrical',
'electrical',
'electrical-',
'electrical/',
'electrical/IT',
'electrical/data',
'electrical/dry',
'electrical/gas',
'electrical/lighting',
'electrical/plumb/mech',
'electrical/plumbing',
'electrical/plumbing/mechanical',
'electrical/plumbing_x000D_',
'electrical2',
'electrical_x000D_',
'electrically',
'electricalpanel',
'electricals',
'electrician',
'electricity',
'electricity/plumbing',
'electricl',
'electricutility',
'framing/electrical',
'gas-electric',
'gas/electric',
'gas/electrical',
'gas/electrics',
'heatpump/electric',
'lighting/electrical',
'mechanical/electrical',
'mechanical/electrical/plumbing',
'non-electrical',
'orelectrical',
'pergola-electrical',
'plumbing/electrical',
'plumbing/mech/electric',
'propane/electric',
'solar/electrical',
'w/electric',
'w/electrical',
'w/electrical-1',
'w/electrical_x000D_']
```

```
In [52]: #clean up useless words
print("Before:" , len(flat_alldes))
print("After:" , len(clean_flat))
```

Before: 4889373  
After: 3534821

```
In [59]: Counter(clean_flat).most_common(30)
```

```
Out[59]: [('PV', 41580),
 ('SOLAR', 40698),
 ('New', 34200),
 ('Remodel', 33179),
 ('AND', 32540),
 ('ROOF', 30209),
 ('NEW', 27167),
 ('SYSTEM', 26579),
 ('TO', 26508),
 ('existing', 25424),
 ('Home', 24849),
 ('Residential', 21690),
 ('unit', 21391),
 ('new', 19850),
 ('KW', 19461),
 ('Replace', 18919),
 ('SF', 18914),
 ('WITH', 17738),
 ('Solar', 17602),
 ('OF', 16953),
 ('system', 15507),
 ('Install', 15178),
 ('', 14991),
 ('Addition', 14668),
 ('INSTALL', 14568),
 ('FOR', 14543),
 ('Roof', 14098),
 ('MODULES', 14072),
 ('System', 13758),
 ('RESIDENTIAL', 13672)]
```

**\*\*\*Count Prescriptive Compliance - heat pump in climate zone 3, 4, 13, and 14 for all projects issued after 1/1/2023\*\*\***

**OLD /\* IGNORE \*/**

```
In [5]: df[df['CODE*'] == 'U434']['DESC'].unique()
df['UCODE_TYPE'].unique()
```

```
Out[5]: array(['New Residential Buildings', 'New Non-Residential Buildings',
 'Additions, Alterations and Conversions of Existing Structures',
 'Energy Permits'], dtype=object)
```

```

In [6]: #df[df['CODE*'] == 'U434'].head()
# Create specific dataframes for building populations: Res, Nonres, MF, Pr
# New Construction
print(f"These are the original Counts:")
df_new_res = df.loc[df['CODE*'].isin(['U101', 'U102', 'U103'])]
print(f"This is the count of New res permits: {len(df_new_res)}")
#print(df_new_res.head())
df_new_MF = df.loc[df['CODE*'].isin(['U104', 'U105'])]
print(f"This is the count of new MF permits: {len(df_new_MF)}")
df_new_nonres = df.loc[df['CODE*'].isin(['U213', 'U214', 'U318', 'U319', 'U321'])]
print(f"This is the count of new Nonres permits: {len(df_new_nonres)}")
# Additions, Alterations, and Conversions of Existing Structures
df_res_addalt = df.loc[df['CODE*'].isin(['U434', 'U434HVAC', 'U438'])]
print(f"This is the count of res add/alt permits: {len(df_res_addalt)}")
df_nonres_addalt = df.loc[df['CODE*'].isin(['U437', 'U437HVAC'])]
print(f"This is the count of nonres add/alt permits: {len(df_nonres_addalt)}")
# Industrial Processes and Energy Permits
df_process = df.loc[df['CODE*'].isin(['U320'])]
print(f"This is the count of industrial process permits: {len(df_process)}")
df_res_energy_permits = df.loc[df['CODE*'].isin(['U340R', 'U340NR', 'U500R'])]
print(f"This is the count of res energy permits: {len(df_res_energy_permits)}")
df_nonres_energy_permits = df.loc[df['CODE*'].isin(['U340C', 'U500C'])]
print(f"This is the count of nonres energy permits: {len(df_nonres_energy_permits)}")
df_EV_energy_permits = df.loc[df['CODE*'].isin(['U502'])]
print(f"This is the count of EV energy permits: {len(df_EV_energy_permits)}")
# Below are QA/QC Checks of the permit counts
#print(f"This is the count of the above permits: {len(df_new_res) + len(df_new_MF) + len(df_new_nonres) + len(df_res_addalt) + len(df_nonres_addalt) + len(df_process) + len(df_res_energy_permits) + len(df_nonres_energy_permits) + len(df_EV_energy_permits)}")
#print(f"This is the count of all permits: {len(df)}")

```

These are the original Counts:  
 This is the count of New res permits: 54028  
 This is the count of new MF permits: 2542  
 This is the count of new Nonres permits: 42241  
 This is the count of res add/alt permits: 115675  
 This is the count of nonres add/alt permits: 22322  
 This is the count of industrial process permits: 137  
 This is the count of res energy permits: 126056  
 This is the count of nonres energy permits: 2248  
 This is the count of EV energy permits: 9515

```
In [7]: print(f"These are the new NonResidential counts:")
df_new_nonres = df.loc[df['CODE*'].isin(['U213', 'U214', 'U318', 'U319', 'U321'])]
print(f"This is the count of new Nonres permits: {len(df_new_nonres)}")
df_new_nonres_addalt = df.loc[df['CODE*'].isin(['U437'])]
print(f"This is the count of nonres add/alt permits: {len(df_new_nonres_addalt)}")
df_process = df.loc[df['CODE*'].isin(['U320'])]
print(f"This is the count of industrial process permits: {len(df_process)}")
df_other_structures = df.loc[df['CODE*'].isin(['U329'])]
print(f"This is the count of all other permitted structures: {len(df_other_structures)}")

print(f"These are the new Residential counts:")
df_new_SFH = df.loc[df['CODE*'].isin(['U101'])]
print(f"This is the count of New SFH permits: {len(df_new_SFH)}")
df_new_ADU = df.loc[df['CODE*'].isin(['U102'])]
print(f"This is the count of New ADU permits: {len(df_new_ADU)}")
df_new_2_4MF = df.loc[df['CODE*'].isin(['U103', 'U104'])]
print(f"This is the count of New 2 to 4 unit permits: {len(df_new_2_4MF)}")
df_new_5MF = df.loc[df['CODE*'].isin(['U105'])]
print(f"This is the count of New 5+ unit permits: {len(df_new_5MF)}")
df_newalt_garage = df.loc[df['CODE*'].isin(['U438'])]
print(f"This is the count of New/Alt garage permits: {len(df_newalt_garage)}")
df_new_res_addalt = df.loc[df['CODE*'].isin(['U434'])]
print(f"This is the count of residential alt/add permits: {len(df_new_res_addalt)}")

print(f"These are the Energy Permit counts:")
df_new_res_energy_permits = df.loc[df['CODE*'].isin(['U340R', 'U340NR'])]
print(f"This is the count of res energy permits: {len(df_new_res_energy_permits)}")
df_new_nonres_energy_permits = df.loc[df['CODE*'].isin(['U340C'])]
print(f"This is the count of nonres energy permits: {len(df_new_nonres_energy_permits)}")
df_EV_energy_permits = df.loc[df['CODE*'].isin(['U502'])]
print(f"This is the count of EV energy permits: {len(df_EV_energy_permits)}")
df_battery_energy_permits = df.loc[df['CODE*'].isin(['U503'])]
print(f"This is the count of Battery energy permits: {len(df_battery_energy_permits)}")

print(f"These are the Mechanical Permit counts:")
df_res_HVAC = df.loc[df['CODE*'].isin(['U434HVAC', 'U500R'])]
print(f"This is the count of res HVAC permits: {len(df_res_HVAC)}")
df_nonres_HVAC = df.loc[df['CODE*'].isin(['U437HVAC', 'U500C'])]
print(f"This is the count of nonres HVAC permits: {len(df_nonres_HVAC)}")
df_gen_mech = df.loc[df['CODE*'].isin(['U501'])]
print(f"This is the count of general mechanical permits: {len(df_gen_mech)}")
df_plumb_mech = df.loc[df['CODE*'].isin(['U600'])]
print(f"This is the count of plumbing mechanical permits: {len(df_plumb_mech)}")

print(f"Below are QA/QC Checks of the permit counts:")

print(f"This is the count of the residential permits: {len(df_new_SFH) + len(df_new_ADU) + len(df_new_2_4MF) + len(df_new_5MF) + len(df_newalt_garage) + len(df_new_res_addalt)}")
print(f"This is the count of the nonresidential permits: {len(df_new_nonres) + len(df_new_nonres_addalt) + len(df_process) + len(df_other_structures)}")
print(f"This is the count of the energy permits: {len(df_new_res_energy_permits) + len(df_new_nonres_energy_permits) + len(df_EV_energy_permits) + len(df_battery_energy_permits)}")
print(f"This is the count of the mechanical permits: {len(df_res_HVAC) + len(df_nonres_HVAC) + len(df_gen_mech) + len(df_plumb_mech)}")

print(f"This is the count of all permits: {len(df)}")
```

These are the new NonResidential counts:  
This is the count of new Nonres permits: 7416  
This is the count of nonres add/alt permits: 21979  
This is the count of industrial proccess permits: 137  
This is the count of all other permitted structures: 34825  
These are the new Residential counts:  
This is the count of New SFH permits: 40104  
This is the count of New ADU permits: 12853  
This is the count of New 2 to 4 unit permits: 1667  
This is the count of New 5+ unit permits: 1946  
This is the count of New/Alt garage permits: 3405  
This is the count of residential alt/add permits: 99862  
These are the Energy Permit counts:  
This is the count of res energy permits: 95554  
This is the count of nonres energy permits: 952  
This is the count of EV energy permits: 9515  
This is the count of Battery energy permits: 0  
These are the Mechanical Permit counts:  
This is the count of res HVAC permits: 42910  
This is the count of nonres HVAC permits: 1639  
This is the count of general mechanical permits: 0  
This is the count of plumbing mechanical permits: 0  
Below are QA/QC Checks of the permit counts:  
This is the count of the residential permits: 159837  
This is the count of the nonresidential permits: 64357  
This is the count of the energy permits: 106021  
This is the count of the mechanical permits: 44549  
This is the count of all permits: 374764

```
In [8]: pd.DataFrame({'UNITS' : df.groupby('CODE*')['VALUATION*'].sum()}).reset_in
```

Out[8]:

|    | CODE*    | UNITS        |
|----|----------|--------------|
| 0  | U101     | 1.402996e+10 |
| 1  | U102     | 1.730089e+09 |
| 2  | U103     | 4.460401e+08 |
| 3  | U104     | 3.507716e+08 |
| 4  | U105     | 8.430590e+09 |
| 5  | U213     | 4.773785e+08 |
| 6  | U214     | 4.040748e+07 |
| 7  | U318     | 1.277806e+08 |
| 8  | U319     | 9.825276e+07 |
| 9  | U320     | 1.049295e+09 |
| 10 | U321     | 1.203951e+09 |
| 11 | U322     | 9.870626e+06 |
| 12 | U323     | 2.574596e+08 |
| 13 | U324     | 5.078456e+08 |
| 14 | U325     | 7.540150e+08 |
| 15 | U326     | 8.485597e+07 |
| 16 | U327     | 3.135594e+09 |
| 17 | U328     | 2.416770e+09 |
| 18 | U329     | 1.686719e+09 |
| 19 | U340C    | 6.983371e+08 |
| 20 | U340NR   | 1.319258e+07 |
| 21 | U340R    | 1.902163e+09 |
| 22 | U434     | 5.896085e+09 |
| 23 | U434HVAC | 1.126815e+08 |
| 24 | U437     | 9.675408e+09 |
| 25 | U437HVAC | 2.596254e+07 |
| 26 | U438     | 1.818641e+08 |
| 27 | U500C    | 1.249721e+08 |
| 28 | U500R    | 3.402834e+08 |
| 29 | U502     | 1.476601e+08 |

```
In [9]: df.isnull().sum()
```

```
Out[9]: COUNTY*          0
        CITY*           0
        MONTH*          0
        YEAR*           0
        ISSUE DATE*      0
        PERMIT NO        0
        SOURCE*          0
        CODE*            0
        CLASS*           0
        APN              15
        ADDRESS          0
        CITY             0
        STATE            0
        ZIP              0
        BUILDER          12
        SQ FOOTAGE       0
        UNITS            0
        VALUATION*       0
        NAME OF OWNER    1
        PERMIT COST      0
        NUMBER OF STORIES 0
        CAPACITY         0
        DESC             75
        CODE             0
        UCODE_TYPE       0
        UCODE_BLG_CAT    0
        UCODE_NAME       0
        UCODE_DESC       0
        dtype: int64
```

```
In [10]: df['UCODE_BLG_CAT'].unique()
```

```
Out[10]: array(['Residential', 'Non-residential', 'Both'], dtype=object)
```

```
In [11]: df.groupby(['UCODE_TYPE', 'UCODE_BLG_CAT']).count()
```

Out[11]:

|                                                                              |                 | COUNTY* | CITY*  | MONTH* | YEAR*  | ISSUE<br>DATE* | PERMIT<br>NO | SC |
|------------------------------------------------------------------------------|-----------------|---------|--------|--------|--------|----------------|--------------|----|
| UCODE_TYPE                                                                   | UCODE_BLG_CAT   |         |        |        |        |                |              |    |
| Additions,<br>Alterations<br>and<br>Conversions<br>of Existing<br>Structures | Non-residential | 22322   | 22322  | 22322  | 22322  | 22322          | 22322        |    |
|                                                                              | Residential     | 115675  | 115675 | 115675 | 115675 | 115675         | 115675       |    |
| Energy<br>Permits                                                            | Both            | 9515    | 9515   | 9515   | 9515   | 9515           | 9515         |    |
|                                                                              | Non-residential | 2248    | 2248   | 2248   | 2248   | 2248           | 2248         |    |
|                                                                              | Residential     | 126056  | 126056 | 126056 | 126056 | 126056         | 126056       |    |
| New Non-<br>Residential<br>Buildings                                         | Non-residential | 42378   | 42378  | 42378  | 42378  | 42378          | 42378        |    |
| New<br>Residential<br>Buildings                                              | Residential     | 56570   | 56570  | 56570  | 56570  | 56570          | 56570        |    |

7 rows × 26 columns

```
In [12]: df[(df['UCODE_TYPE'] == 'Additions, Alterations and Conversions of Existin
```

Out[12]:

|        | COUNTY*      | CITY*                       | MONTH*    | YEAR*  | ISSUE DATE* | PERMIT NO | SOURCE    |
|--------|--------------|-----------------------------|-----------|--------|-------------|-----------|-----------|
| 184978 | BUTTE        | BUTTE COUNTY UNINCORPORATED | JANUARY   | 2023.0 | 1/27/2023   | B22-4191  | SECONDARY |
| 184979 | BUTTE        | BUTTE COUNTY UNINCORPORATED | JANUARY   | 2023.0 | 1/19/2023   | B22-3707  | SECONDARY |
| 184980 | BUTTE        | BUTTE COUNTY UNINCORPORATED | MARCH     | 2023.0 | 3/16/2023   | B22-2556  | SECONDARY |
| 184981 | BUTTE        | BUTTE COUNTY UNINCORPORATED | APRIL     | 2023.0 | 4/11/2023   | B23-0186  | SECONDARY |
| 184982 | BUTTE        | BUTTE COUNTY UNINCORPORATED | APRIL     | 2023.0 | 4/17/2023   | B22-3256  | SECONDARY |
| ...    | ...          | ...                         | ...       | ...    | ...         | ...       | ...       |
| 370171 | ALAMEDA      | PLEASANTON                  | AUGUST    | 2023.0 | 8/2/2023    | B23-2884  | PRIMARY   |
| 370172 | ALAMEDA      | PLEASANTON                  | AUGUST    | 2023.0 | 8/31/2023   | B23-2522  | PRIMARY   |
| 370173 | ALAMEDA      | PLEASANTON                  | SEPTEMBER | 2023.0 | 9/26/2023   | B23-2913  | PRIMARY   |
| 370174 | CONTRA COSTA | CONCORD                     | FEBRUARY  | 2023.0 | 2/20/2023   | B211071   | PRIMARY   |
| 370175 | CONTRA COSTA | CONCORD                     | MARCH     | 2023.0 | 3/3/2023    | B230177   | PRIMARY   |

22322 rows x 28 columns

```
In [ ]: WordCloud().generate(str(existing_nonres_text))
```

### **Geomap in Python**

```
In [ ]: pd.set_option('display.max_columns', None)
#pd.set_option('display.max_row', None)
pd.set_option('display.max_colwidth', None)
```

```
In [ ]: #clean up the df so that it has the category type by zip and count

#geocode with test df
df_temp = df.iloc[1:150]
```

```
In [ ]: #df_temp.nunique()
```

```
In [ ]: #see duplicated address
#df_temp[df_temp.duplicated('ADDRESS', keep=False) == True]
```

```
In [ ]: # Initialize Nominatim API for the US
nomi = pgeocode.Nominatim('us')

#Get Long Lat with zipcode
def get_long(x):
    location = nomi.query_postal_code(x)
    return location.longitude

def get_lat(x):
    location = nomi.query_postal_code(x)
    return location.latitude
```

```
In [ ]: %%time

#get Long and Lat
#df['Long'] = df['ZIP'].apply(lambda x: get_long(x))
#df['Lat'] = df['ZIP'].apply(lambda x: get_lat(x))
```

```
In [ ]: #df_temp.head()
```

```
In [ ]: df_count = df.groupby('COUNTY*').APN.nunique()
df_count = pd.DataFrame(df_count)
df_count = df_count.reset_index()
df_count.head()
```

```
In [ ]: #df['Lat'].isnull().sum()/len(df)
```

```
In [ ]: #import shape map
ca_state = geopandas.read_file('GeoData/ca_zipcodes/California_Zip_Codes.s
ca_state_bycounty = geopandas.read_file('GeoData/ca_counties/CA_Counties.s
type(ca_state)
```

```
In [ ]: ca_state_bycounty.crs
```

```
In [ ]: ca_state_county = ca_state_bycounty.to_crs("epsg:4326")
```

```
In [ ]: crs = {'init':'epsg:4326'}
```

```
In [ ]: len(ca_state_county)
```

```
In [ ]: ca_state_county['NAME'] = ca_state_county['NAME'].str.upper()  
test = pd.merge(ca_state_county,df_count, how='left',  
                left_on='NAME',right_on='COUNTY*')
```

```
In [ ]: #test  
print(test.columns.tolist())
```

```
In [ ]: df_count.head()
```

```
In [ ]: df_count = df_temp.groupby(["CODE*", "ZIP"], as_index=False).size()  
df_count['Long'] = df_count['ZIP'].apply(lambda x: get_long(x))  
df_count['Lat'] = df_count['ZIP'].apply(lambda x: get_lat(x))  
df_count.head()
```

```
In [ ]: #map  
geometry = [Point(xy) for xy in zip(df_count['Long'], df_count['Lat'])]  
#geometry[:3]  
gdf = GeoDataFrame(df_count, crs = crs, geometry=geometry)
```

```
In [ ]: df_count['Lat'].isnull().sum()/len(df_count)
```

```
In [ ]: world_data = geopandas.read_file(geopandas.datasets.get_path('naturalearth'))  
df_count.plot(ax=world_data[world_data.continent == 'North America'].plot()  
plt.savefig('world.jpg')
```

```
ca_state_bycounty.plot(  
    column="POPULATION",  
    legend=True,  
    legend_kwds = {"label":"Population", "orientation":"horizontal"})
```

```
In [ ]: fig, ax = plt.subplots(figsize=(10,10))  
ca_state_county.plot(ax = ax, alpha = 0.4)  
gdf.plot(ax = ax, markersize=10, color='red', marker='o')  
plt.legend(prop={'size':15})
```

```
In [ ]: fig, ax = plt.subplots(figsize=(10,10))  
test.plot(  
    column="APN",  
    missing_kwds={'color':'lightgrey'},  
    ax = ax,  
    legend=True,  
    legend_kwds={'label':'Number of Permits by County', 'orientation':'ver
```

In [ ]: